
Supplement: Characterizing Perfect Truthfulness in Continuous Online Calibration

Pahan Dewasurendra
Johns Hopkins University

A Probability Trees and Proper Scores

Let $\mathcal{A} = [k]$ and let Δ_k be the probability simplex. For depth d , define the report-tree space

$$\mathcal{R}_d = \prod_{t=1}^d \prod_{h \in \mathcal{A}^{t-1}} \Delta_k.$$

We write $r_h \in \Delta_k$ for the report at history h . The report tree $R \in \mathcal{R}_d$ induces a law Q_R on $\mathcal{A}^d$ by

$$Q_R(x) = \prod_{t=1}^d r_{x_{<t}, x_t}. \quad (1)$$

On the relative interior $\mathcal{R}_d^\circ$, this is a bijection between report trees and full-support laws on $\mathcal{A}^d$. Its inverse takes the usual conditional probabilities. Both maps are continuous on the relative interiors.

A realized-path score has the form

$$S(x, R) = C(x, (r_{x_{<t}})_{t=1}^d). \quad (2)$$

Perfect truthfulness means that S is a proper score for the joint law: for every law P and every report tree R ,

$$\mathbb{E}_P S(X, R_P) \leq \mathbb{E}_P S(X, R), \quad (3)$$

where R_P is any conditional-probability tree representing P . Reports at histories outside the support of P do not affect either side. On full-support laws, R_P is unique.

We use losses, so lower is better. For a finite outcome space Ω , a proper loss is a continuous vector map $s : \Delta(\Omega) \rightarrow \mathbb{R}^\Omega$ such that

$$p \cdot s(p) \leq p \cdot s(q) \quad \text{for all } p, q \in \Delta(\Omega). \quad (4)$$

Its entropy or Bayes risk is $H_s(p) = p \cdot s(p)$.

B Continuous Proper-Score Lemmas

For completeness, we prove the two convex-analytic lemmas used in the main paper. They hold on any finite outcome space and require only continuity.

Lemma 1 (Continuous support). *Let $s : \Delta(\Omega) \rightarrow \mathbb{R}^\Omega$ be continuous and proper. For every relative-interior p and every tangent vector v satisfying $\mathbf{1} \cdot v = 0$,*

$$DH_s(p)[v] = v \cdot s(p). \quad (5)$$

The derivative is continuous in p .

Proof. Choose $t > 0$ small enough that $p + tv$ remains in the relative interior. Propriety at truth $p + tv$ with report p gives

$$H_s(p + tv) \leq (p + tv) \cdot s(p) = H_s(p) + tv \cdot s(p).$$

Propriety at truth p with report $p + tv$ gives

$$\begin{aligned} H_s(p) &\leq p \cdot s(p + tv) \\ &= H_s(p + tv) - tv \cdot s(p + tv). \end{aligned}$$

Therefore

$$v \cdot s(p + tv) \leq \frac{H_s(p + tv) - H_s(p)}{t} \leq v \cdot s(p). \quad (6)$$

Letting $t \downarrow 0$ and using continuity proves the right directional derivative. Replacing v by $-v$ proves the left derivative. The expression $v \cdot s(p)$ is continuous in p . $\square$

Corollary 2 (Entropy uniqueness). *Let s and u be continuous proper losses on a finite outcome space. If $H_s(p) - H_u(p) = c$ for every relative-interior p , then*

$$s_i(p) - u_i(p) = c \quad \text{for every } i, p. \quad (7)$$

The conclusion extends to the simplex boundary. The componentwise conclusion on the relative interior requires only that the scores be continuous and proper when truths and reports are restricted to the relative interior.

Proof. Lemma 1 gives $v \cdot (s(p) - u(p)) = 0$ for every tangent vector v . Hence $s(p) - u(p) = \lambda(p)\mathbf{1}$. Taking the inner product with p shows $\lambda(p) = c$. Continuity extends the identity to the boundary. $\square$

Lemma 3 (Shared component). *Let s and u be continuous proper losses on Δ_k . Fix $i \in [k]$. If*

$$s_j(q) = u_j(q) \quad \text{for every } j \neq i \text{ and } q \in \Delta_k,$$

then $s_i(q) - u_i(q)$ is constant in q .

Proof. Work first in the relative interior. Set $d(q) = s_i(q) - u_i(q)$ and

$$F(q) = H_s(q) - H_u(q) = q_i d(q).$$

By Lemma 1, for every tangent vector v ,

$$DF(q)[v] = v \cdot (s(q) - u(q)) = v_i d(q). \quad (8)$$

If $v_i = 0$, the derivative is zero. Each relative-interior slice with fixed $q_i = z$ is connected, so $F(q) = f(z)$ for some continuously differentiable function $f : (0, 1) \rightarrow \mathbb{R}$.

Choose any $j \neq i$ and move in direction $e_i - e_j$. Equation (8) gives

$$f'(z) = d(q) = \frac{f(z)}{z}.$$

Thus $(f(z)/z)' = 0$, so $f(z) = cz$ and $d(q) = c$. Continuity extends the result to the boundary. $\square$

C Proof of the Path-Score Characterization

We restate the representation in report-tree notation.

Theorem 4 (Path-score characterization). *Let C be a continuous realized-path score of depth d on a finite alphabet. It is proper for every law on $\mathcal{A}^d$ if and only if there are continuous proper one-step losses ℓ_h such that*

$$C(x, (r_{x_{<t}})_t) = \sum_{t=1}^d \ell_{x_{<t}}(x_t, r_{x_{<t}}). \quad (9)$$

Proof. The reverse implication follows from iterated conditioning and one-step propriety. We prove the forward implication by induction on d . The claim is immediate for $d = 1$.

Assume the claim through depth $d - 1$ and consider a depth- d score. We first work with interior root reports and full-support continuation laws. Fix an interior root report $q \in \Delta_k^\circ$. For root outcome i , define the continuation score

$$C_i^q(z, R) = C(iz, (q, (r_{z_{<t}})_t)), \quad (10)$$

where $z \in \mathcal{A}^{d-1}$ and $R \in \mathcal{R}_{d-1}$. The index alignment in (10) is only notational: R supplies every report after the root on branch i .

For fixed q , C_i^q is proper for every continuation law P_i . To verify this, construct a joint law whose truthful root report is q , whose branch i continuation is P_i , and whose other continuations are arbitrary and full support. Compare the truthful tree with a tree that changes only the continuation reports in branch i . Joint propriety gives the continuation properness inequality multiplied by $q_i > 0$.

Let

$$B_i(q, P_i) = \mathbb{E}_{Z \sim P_i} C_i^q(Z, R_{P_i}) \quad (11)$$

be the continuation Bayes risk. Fix a tuple of full-support continuation laws $P = (P_1, \dots, P_k)$. The vector

$$B(q, P) = (B_1(q, P_1), \dots, B_k(q, P_k))$$

is a continuous proper root loss. Indeed, for a true root law p , compare the fully truthful report tree with a tree that reports q at the root and reports every continuation P_i truthfully. Joint propriety gives

$$\sum_i p_i B_i(p, P_i) \leq \sum_i p_i B_i(q, P_i).$$

Fix reference continuation laws P_i^0 and an interior reference root report q^0 . If P_i changes while every other continuation stays fixed, the two proper root losses agree in every component except i . By Lemma 3,

$$B_i(q, P_i) - B_i(q, P_i^0)$$

is independent of q . Define

$$\begin{aligned} A_i(q) &= B_i(q, P_i^0), \\ K_i(P_i) &= B_i(q^0, P_i) - B_i(q^0, P_i^0). \end{aligned} \quad (12)$$

Then

$$B_i(q, P_i) = A_i(q) + K_i(P_i). \quad (13)$$

The vector $A(q) = (A_i(q))_{i=1}^k$ is a proper root loss. For any fixed continuation tuple P , the proper loss $B(q, P)$ differs from $A(q)$ by the outcome-dependent constant vector $(K_i(P_i))_i$. Such constants cancel from the properness comparison at a fixed true root law.

It remains to separate the full continuation score, not only its Bayes risk. For fixed i and root reports q, q^0 , both C_i^q and $C_i^{q^0}$ are continuous proper scores on the finite outcome space $\mathcal{A}^{d-1}$. Under the bijection in (1), their entropies at a full-support continuation law P_i are $B_i(q, P_i)$ and $B_i(q^0, P_i)$. By (13), their difference is the constant $A_i(q) - A_i(q^0)$. The interior version of Corollary 2 gives

$$C_i^q(z, R) = C_i^{q^0}(z, R) + A_i(q) - A_i(q^0) \quad (14)$$

for every full-support continuation report tree R and every z .

Define

$$D_i(z, R) = C_i^{q^0}(z, R) - A_i(q^0).$$

Then $C_i^q(z, R) = A_i(q) + D_i(z, R)$. The score D_i is continuous, realized-path local, and proper on every depth- $(d-1)$ continuation law. Apply the induction hypothesis to each D_i . Together with the proper root loss A , this gives (9) for interior reports.

Interior report trees are dense in $\mathcal{R}_d$. Every term and the original score are continuous, so the identity extends to all boundary reports. The properness inequalities for the component losses also extend to boundary true laws and reports by continuity. $\square$

D Normalization and Risk Comparison

Corollary 5 (Canonical normalization). *Suppose the score in Theorem 4 satisfies*

$$C(x, (e_{x_t})_{t=1}^d) = 0 \quad \text{for every } x \in \mathcal{A}^d. \quad (15)$$

Then the components can be chosen to satisfy

$$\ell_h(i, e_i) = 0, \quad \ell_h(i, q) \geq 0.$$

Proof. Starting from any representation, define

$$\tilde{\ell}_h(i, q) = \ell_h(i, q) - \ell_h(i, e_i).$$

Subtracting an outcome-dependent constant preserves propriety. Propriety at the deterministic truth e_i gives $\ell_h(i, e_i) \leq \ell_h(i, q)$, so every normalized loss is nonnegative. Finally, for each path x ,

$$\sum_{t=1}^d \ell_{x_{<t}}(x_t, e_{x_t}) = C(x, (e_{x_t})_t) = 0.$$

Thus replacing every component by its normalized version leaves C exactly unchanged. $\square$

Lemma 6 (One-step comparison). *Let ℓ be a nonnegative proper categorical loss normalized at every simplex vertex. For interior $p, q \in \Delta_k$, define*

$$H(p) = \sum_i p_i \ell(i, p),$$

$$L(p, q) = \sum_i p_i \ell(i, q).$$

Then

$$L(p, q) \leq \left(\max_i \frac{p_i}{q_i} \right) \left(\max_i \frac{q_i}{p_i} \right) H(p). \quad (16)$$

Proof. Nonnegativity gives

$$L(p, q) \leq \left(\max_i \frac{p_i}{q_i} \right) \sum_i q_i \ell(i, q).$$

Properness at truth q gives

$$\sum_i q_i \ell(i, q) \leq \sum_i q_i \ell(i, p).$$

A second use of nonnegativity gives

$$\sum_i q_i \ell(i, p) \leq \left(\max_i \frac{q_i}{p_i} \right) H(p).$$

Multiplying the factors proves the claim. $\square$

Theorem 7 (Sequential comparison). *Let C_T be continuous, perfectly truthful, and satisfy (15). For interior $p, q \in \Delta_k$ and i.i.d. outcomes $X_t \sim p$,*

$$\mathbb{E}_p C_T(X, q \mathbf{1}_T) \leq \kappa(p, q) \mathbb{E}_p C_T(X, p \mathbf{1}_T), \quad (17)$$

where

$$\kappa(p, q) = \left(\max_i \frac{p_i}{q_i} \right) \left(\max_i \frac{q_i}{p_i} \right).$$

Proof. Use the normalized components from Corollary 5. At each history h , Lemma 6 applies to ℓ_h . Under i.i.d. truth p , the conditional law of X_t given $X_{<t} = h$ is p . Therefore

$$\begin{aligned} \mathbb{E}_p[\ell_{X_{<t}}(X_t, q) \mid X_{<t}] \\ \leq \kappa(p, q) \mathbb{E}_p[\ell_{X_{<t}}(X_t, p) \mid X_{<t}]. \end{aligned}$$

Take expectations and sum over t . $\square$

For binary $p = (1-a, a)$ and $q = (1-b, b)$, if $b > a$ then

$$\begin{aligned} \max_i \frac{p_i}{q_i} &= \frac{1-a}{1-b}, \\ \max_i \frac{q_i}{p_i} &= \frac{b}{a}. \end{aligned}$$

Their product is $b(1-a)/(a(1-b))$. The case $b < a$ gives its reciprocal. Thus

$$\kappa(p, q) = \exp(|\logit(a) - \logit(b)|). \quad (18)$$

E Sharpness of the Binary Constant

We give a direct construction rather than invoke the full representation theory of binary proper losses. Let $w : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$ be a continuous density. Define

$$\ell(0, q) = \int_0^q tw(t) dt, \quad (19)$$

$$\ell(1, q) = \int_q^1 (1-t)w(t) dt. \quad (20)$$

The losses are continuous, nonnegative, and normalized at $q = 0$ and $q = 1$. For truth a , differentiation of the cross-risk gives

$$\frac{d}{dq} ((1-a)\ell(0, q) + a\ell(1, q)) = (q-a)w(q).$$

Hence the loss is proper.

Fix $b > a$. Choose a sequence of continuous unit-mass densities w_n supported inside (a, b) and concentrating at b from below. Since all mass lies above a and below b ,

$$L_n(a, a) = a \int_a^b (1-t)w_n(t) dt,$$

$$L_n(a, b) = (1-a) \int_a^b tw_n(t) dt.$$

Therefore

$$\frac{L_n(a, b)}{L_n(a, a)} \rightarrow \frac{b(1-a)}{a(1-b)} = \kappa(p, q).$$

For $b < a$, choose densities supported inside (b, a) and concentrating at b from above. Then

$$L_n(a, a) = (1-a) \int_b^a tw_n(t) dt,$$

$$L_n(a, b) = a \int_b^a (1-t)w_n(t) dt,$$

and the ratio approaches $a(1-b)/(b(1-a))$. This proves the sharpness claim for every $\epsilon > 0$.

F Batch ATB and Its Online Deviation

For completeness, define the empirical Averaged Two-Bin calibration error of Hartline et al. (2026) by

$$\text{ATB}_T(r, x) = \frac{1}{T^2} \int_0^1 \left[\left(\sum_{t:r_t < z} (r_t - x_t) \right)^2 + \left(\sum_{t:r_t \geq z} (r_t - x_t) \right)^2 \right] dz. \quad (21)$$

The choice of strict inequality at the threshold is immaterial under the integral.

If $r = b\mathbf{1}_T$, all reports occupy one bin for almost every z . Hence

$$\text{ATB}_T(b\mathbf{1}_T, x) = \frac{1}{T^2} \left(\sum_{t=1}^T (b - x_t) \right)^2.$$

For i.i.d. $X_t \sim \text{Ber}(a)$, bias and variance give

$$\mathbb{E}_a \text{ATB}_T(b\mathbf{1}_T, X) = (b-a)^2 + \frac{a(1-a)}{T}. \quad (22)$$

Setting $b = a$ yields truthful expected ATB $a(1-a)/T$.

We next verify the two-round online deviation in the main paper. Let $a = 1/2$, set $R_1 = 1/2$, and set $R_2 = 1/4$ after $X_1 = 0$ or $R_2 = 3/4$ after $X_1 = 1$. Integrating (21) over the intervals induced by the two reports gives the following table.

$X_1 X_2$	(R_1, R_2)	ATB ₂
00	(1/2, 1/4)	1/8
01	(1/2, 1/4)	1/16
10	(1/2, 3/4)	1/16
11	(1/2, 3/4)	1/8

The outcomes are equiprobable, so the adaptive expected loss is

$$\frac{1}{4} \left(\frac{1}{8} + \frac{1}{16} + \frac{1}{16} + \frac{1}{8} \right) = \frac{3}{32}.$$

For truthful constant reports, (22) with $T = 2$ gives $1/8$. Thus the adaptive reports strictly improve on truth.

References

Hartline, J., Hu, L., and Wu, Y. (2026). A perfectly truthful calibration measure. In *Proceedings of the 39th Annual Conference on Learning Theory*, volume 336 of *Proceedings of Machine Learning Research*, pages 3185–3223. PMLR.